PROGRAM 2 – 8-QUEEN PROBLEM

AIM

To write a Python program to solve the 8-Queen problem using backtracking.

ALGORITHM

1. Start from the first row.
2. Place one queen in each row.
3. Check whether the selected position is safe.
4. Check column and diagonal conflicts.
5. If safe, place the queen.
6. Move to the next row.
7. If no safe position is available, backtrack.
8. Continue until all 8 queens are placed.

CODE

```
N = 8

board = [-1] * N



def is_safe(row, col):
    for i in range(row):

        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True



def solve(row):
    if row == N:
```

```
            return True

        for col in range(N):

            if is_safe(row, col):
                board[row] = col

                if solve(row + 1):
                    return True

                board[row] = -1

        return False



if solve(0):

    print("Solution:")

    for row in range(N):

        for col in range(N):

            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")

        print()
```

INPUT:

```
2.8_queen.py > ...
1    N = 8
```

OUTPUT:

```
Solution:
Q . . . . . . .
. . . . Q . . .
. . . . . . . Q
. . . . . Q . .
. . Q . . . . .
. . . . . . Q .
. Q . . . . . .
. . . Q . . . .
```

RESULT

Thus, the 8-Queen problem was successfully solved using backtracking.